

Reinforcement Learning Laboratory 2

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Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

Task 1: Identification of Reinforcement Learning Components

RL Component	Observation
Agent	The AI player that learns how to play Tic-Tac-Toe by gaining experience from repeated games.
Environment	The Tic-Tac-Toe board, its rules, and the opponent against whom the agent plays.
State Representation	Every possible arrangement of X, O, and empty spaces on the 3×3 board.
Action Space	Selecting one of the currently unoccupied cells to place the agent's symbol.
Reward Function	A reward of +1 is given for a win, -1 for a loss, and 0 for a draw or while the game is still in progress.
Learning Approach Used	Reinforcement Learning based on self-play, where the agent improves its decisions using feedback from rewards.

1. Who is the learning agent in this environment?

The learning agent is the computer-controlled Tic-Tac-Toe player that improves its gameplay through repeated interactions.

2. What is considered as the environment?

The environment consists of the Tic-Tac-Toe board, the game's rules, and the opponent's actions.

3. How is the game state represented?

The state is represented by the current arrangement of all nine cells, indicating whether each cell contains X, O, or is empty.

4. What are the possible actions available to the agent?

The agent can choose any empty square on the board to place its mark.

5. When does the agent receive a positive reward?

A positive reward is received whenever the agent wins the game.

6. Which Reinforcement Learning approach is used for learning?

The agent uses Reinforcement Learning, learning from repeated gameplay and updating its strategy according to the rewards it receives.

7. Is the agent trained using labelled data? Justify your answer.

No. The agent does not rely on labelled examples. Instead, it learns through experience by trying different actions and receiving rewards based on the outcome of each game.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

Reinforcement Learning	Supervised Learning
Learns by interacting with the game environment	Learns from labelled datasets
Uses rewards to improve decisions	Uses correct answers for training
Improves through trial and error	Improves by minimizing prediction errors
Aims to maximize long-term rewards	Aims to predict correct outputs

Task 2: Effect of Training Episodes on Agent Behaviour

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	31%	51%	18%	Makes mostly random moves and often overlooks winning or blocking opportunities.
500	43%	35%	22%	Begins blocking the opponent and occasionally selects stronger positions.
1,000	57%	20%	23%	Frequently chooses the center and corners while showing better planning.
5,000	69%	7%	24%	Plays with fewer mistakes, blocks consistently, and makes calculated moves.
10,000	73%	3%	24%	Demonstrates near-optimal gameplay, avoids defeats, and frequently forces drawn games.

Behaviour Observed During Training

Behaviour	Approximate Stage
Random gameplay	100 episodes
Starts blocking opponent	500 episodes
Prefers center square	Around 1,000 episodes
Chooses corner cells	Around 1,000 episodes
Makes tactical decisions	Around 5,000 episodes
Minimizes losing	Around 5,000 episodes
Forces draws	Close to 10,000 episodes
Near-optimal gameplay	Around 10,000 episodes

9. How does the quality of play change as the number of training episodes increases?

As the training episodes increase, the agent gradually develops a stronger strategy. It moves from random gameplay to making intelligent decisions, defending effectively, and identifying better opportunities to win.

10. At approximately how many training episodes does the agent begin making intelligent decisions?

The agent begins showing noticeable strategic behaviour after approximately **1,000 training episodes**.

11. At what stage does the learning appear to converge?

The learning process appears to stabilize between **5,000 and 10,000 episodes**, with only small improvements afterward.

12. Why does the win percentage improve with additional training?

With more training, the agent gains experience from a larger number of games. This enables it to recognize better moves, avoid poor decisions, and refine its playing strategy.

13. Why do most games end in a draw after sufficient training?

When both players adopt effective strategies, they successfully block each other's winning chances, making drawn games much more common.

14. What happens if the number of training episodes is too small?

If the training duration is too short, the agent cannot learn effective strategies. As a result, it makes random decisions and loses more frequently.

Conclusion

The experiment demonstrates that reinforcement learning enables the Tic-Tac-Toe agent to improve solely through experience. Initially, the agent relies on random actions and performs poorly. As the number of training episodes increases, it learns to select better positions, defend against the opponent, and make smarter decisions. By around 10,000 training episodes, the agent achieves near-optimal performance, where losses become rare and most games conclude in draws. This illustrates how repeated interaction with the environment allows an RL agent to progressively enhance its policy.